

VS-LEAKKG: A Contamination Knowledge Graph for Leakage-Controlled Virtual Screening Benchmarks

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Abstract

Structure-based virtual screening has increasingly been formulated as a protein–ligand retrieval problem, where protein pockets and candidate ligands are embedded into a shared representation space and ranked by similarity. Despite strong reported performance on benchmarks such as DUD-E, DEKOIS, and LIT-PCBA, the reliability of these results remains difficult to assess due to potential train–test contamination and benchmark-specific shortcuts. Existing evaluation protocols often remove exact target or ligand overlaps, but they rarely account for indirect leakage through ligand scaffolds, homologous proteins, structurally similar pockets, reused assays, shared publications, source-specific decoy protocols, or temporal overlap.

We propose VS-LEAKKG, a contamination-aware heterogeneous graph framework for auditing, constructing, and retraining on leakage-controlled virtual screening benchmarks. VS-LEAKKG represents benchmark examples, proteins, pockets, ligands, scaffolds, assays, publications, dataset sources, decoy protocols, and timestamps as typed nodes, with typed edges encoding possible leakage pathways. The graph enables example-level contamination scoring, multi-axis cold split generation, shortcut diagnostics, and clean train/validation/test construction. We further define a retraining protocol in which modern protein–ligand retrieval models are trained and selected only on leakage-controlled partitions. Our framework therefore provides a benchmark-governance layer for virtual screening, allowing models to be evaluated under progressively stricter and reproducible generalization regimes.

1 Introduction

Structure-based virtual screening (SBVS) is a central task in early-stage drug discovery, aiming to identify active ligands for a target protein from a large candidate library. Recent contrastive protein–ligand models have reformulated SBVS as a retrieval problem, where protein pockets and ligands are embedded into a shared representation space and ranked by similarity. This paradigm enables efficient large-scale screening and has led to strong reported results on established benchmarks such as DUD-E, DEKOIS, and LIT-PCBA.

However, the reliability of these reported gains remains difficult to assess. Many widely used virtual screening benchmarks contain hidden shortcuts that can inflate performance without requiring models to learn true protein–ligand interactions. For instance, DUD-E-style benchmarks construct property-matched but topologically dissimilar decoys, allowing ligand-only models to distinguish actives from decoys by exploiting chemical artifacts rather than binding mechanisms. In addition, modern protein–ligand models are often trained or pretrained on large heterogeneous resources such as ChEMBL, BindingDB, PDDBind, and PubChem-derived datasets. These sources overlap with downstream benchmarks through multiple indirect channels, including shared ligands, similar scaffolds, homologous proteins, structurally similar pockets, reused assays, common publications, and source-specific decoy generation protocols. As a result, a benchmark example may appear “unseen” at the level of an exact protein–ligand pair while still being highly contaminated through its ligand scaffold, protein family, binding pocket, or assay provenance.

Existing efforts address important parts of this problem, but none fully captures the multi-axis contamination structure of virtual screening. DrugOOD studies out-of-distribution generalization for drug–target affinity prediction using domain annotations such as assay, scaffold, protein, and protein family. DataSAIL provides a general-purpose framework for leakage-reduced splitting in biological datasets by minimizing similarity across partitions. Recent leakage-aware protein–ligand benchmarks such as LP-PDBBind and CleanSplit focus on binding affinity or scoring-function evaluation. BayesBind moves closer to SBVS by constructing a benchmark with targets structurally dissimilar from the training set. These works demonstrate the importance of rigorous evaluation, but they do not explicitly model virtual-screening contamination as a typed, high-order relational structure across proteins, pockets, ligands, scaffolds, assays, sources, decoy protocols, and time.

We argue that virtual screening requires a benchmark-governance framework rather than another single benchmark split. In particular, train–test contamination should be treated as a heterogeneous graph problem: examples are connected through biological, chemical, structural, experimental, and provenance relations, and leakage may occur through short paths in this graph. Such a formulation enables three complementary capabilities. First, it enables dataset auditing through example-level and axis-specific contamination scores. Second, it enables construction of train/validation/test splits that explicitly forbid leakage paths across predefined axes. Third, it enables controlled retraining of modern protein–ligand retrieval models under the same leakage-controlled protocol, so that performance reflects generalization rather than overlap with the training data or validation leakage during model selection.

To this end, we propose VS-LEAKKG, a contamination-aware heterogeneous graph framework for auditing and constructing leakage-controlled virtual screening benchmarks. Unlike biomedical knowledge graphs designed for link prediction or drug repurposing, VS-LEAKKG is designed for benchmark integrity: its nodes and edges encode potential leakage pathways between training, validation, and test examples. Using this graph, we quantify contamination along ligand, scaffold, protein, pocket, assay, source, and temporal axes; generate clean multi-axis train/validation/test splits; retrain representative SBVS models on the resulting partitions; and evaluate whether current methods continue to outperform shortcut baselines when contamination is controlled.

2 Contributions

Our main contributions are as follows.

- **A contamination knowledge graph for virtual screening.** We introduce VS-LEAKKG, a typed heterogeneous graph that connects benchmark examples to proteins, protein families, binding pockets, pocket clusters, ligands, molecular scaffolds, assays, publications, dataset sources, decoy protocols, and timestamps. Unlike conventional biomedical knowledge graphs that are primarily used for prediction, VS-LEAKKG is constructed to expose potential train–validation–test leakage pathways in virtual screening benchmarks.
- **Example-level and axis-specific contamination scoring.** We define a path-based contamination score that measures how strongly an example is connected to a reference split through high-confidence leakage paths. The score is decomposed into ligand-level, scaffold-level, protein-level, pocket-level, assay-level, source-level, and temporal components, allowing us to identify not only whether a benchmark is contaminated, but also which type of shortcut dominates.
- **Leakage-controlled train/validation/test split construction.** We propose a graph-based split construction procedure that first collapses examples connected by forbidden leakage paths

into leakage groups, and then partitions these groups into train, validation, and test sets. This ensures that validation and test examples are not connected to the training set through explicitly defined ligand, scaffold, protein, pocket, assay, source, or temporal leakage axes. The resulting splits include ligand-clean, scaffold-clean, protein-clean, pocket-clean, dual-clean, and strict-clean regimes.

- **Model retraining under clean splits.** We define a retraining protocol for modern protein–ligand retrieval models in which model fitting, hyperparameter selection, and final testing are all performed on VS-LEAKKG-clean partitions. This protocol distinguishes leakage in training data from leakage in validation-based checkpoint selection, and allows fair comparison between models under the same cold-start assumptions.
- **Shortcut diagnostics for benchmark validity.** We introduce diagnostic baselines that require little or no protein–ligand interaction modeling, including ligand-only predictors, source-only predictors, contamination-nearest-neighbor predictors, and dummy-receptor controls. These diagnostics test whether high virtual screening performance can be achieved by exploiting chemical, source, or relational artifacts rather than learning structure-based binding signals.
- **A systematic re-evaluation of modern protein–ligand retrieval models.** We apply VS-LEAKKG to audit established and modern virtual screening benchmarks, including DUD-E, LIT-PCBA, and BayesBind-style settings, and re-evaluate representative retrieval-based models such as DrugCLIP, S2Drug, LigUnity, HypSeek, and ConGLUDe. We report standard metrics such as AUROC, BEDROC, and enrichment factors, as well as contamination-aware metrics such as performance by contamination decile, worst-group performance, clean-split performance, and the gap between interaction-based models and shortcut baselines.
- **Evidence that benchmark conclusions depend on contamination control.** Through contamination-aware evaluation and clean-split retraining, we test whether model rankings and absolute performance change when ligand, scaffold, protein, pocket, assay, source, and temporal leakage are controlled simultaneously. These findings motivate measuring progress in virtual screening with benchmark-integrity diagnostics rather than only aggregate enrichment metrics on legacy decoy-based benchmarks.

3 Method

3.1 Problem Formulation

Let a virtual screening corpus consist of a set of examples

$$\mathcal{D} = \{x_i\}_{i=1}^N.$$

Each example is represented as

$$x_i = (p_i, b_i, \ell_i, y_i, s_i),$$

where p_i denotes a target protein, b_i denotes a binding pocket or predicted pocket, ℓ_i denotes a ligand, $y_i \in \{0, 1\}$ denotes the active/inactive label, and s_i denotes metadata such as assay, publication, dataset source, timestamp, and decoy protocol.

A conventional benchmark split partitions the corpus into training, validation, and test subsets:

$$\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{test}}.$$

However, this split may still be contaminated if validation or test examples are connected to the training set through ligand similarity, scaffold identity, protein homology, pocket similarity, assay reuse, shared publications, source-specific decoy protocols, or temporal overlap. Our goal is twofold: (i) to quantify this contamination in existing benchmarks, and (ii) to construct leakage-controlled train/validation/test splits for retraining and evaluating virtual screening models.

3.2 Constructing the Contamination Knowledge Graph

We construct a typed heterogeneous graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R}),$$

where $\mathcal{V}$ denotes nodes, $\mathcal{E}$ denotes edges, and $\mathcal{R}$ denotes edge types. The graph contains multiple node types relevant to virtual screening contamination.

Table 1: Main node types in VS-LEAKKG.

Node type	Description
Example	A benchmark protein–ligand screening example
Protein	Target protein, e.g., UniProt or PDB chain
ProteinCluster	Sequence or family cluster, e.g., 30%, 40%, 90% identity clusters
Pocket	Binding pocket or predicted pocket
PocketCluster	Cluster of structurally or representationally similar pockets
Ligand	Canonical molecule, e.g., InChIKey or canonical SMILES
Scaffold	Molecular scaffold, e.g., Bemis–Murcko scaffold
Assay	Experimental assay from ChEMBL, BindingDB, PubChem, or other sources
Publication	Paper or source record associated with a measurement
DatasetSource	Dataset origin, e.g., DUD-E, LIT-PCBA, PDDBind, ChEMBL
DecoyProtocol	Decoy generation protocol or inactive-labeling process
TimeBin	Timestamp or temporal interval

Edges encode possible leakage pathways. For example, an `example_has_ligand` edge connects an example to its ligand, a `ligand_has_scaffold` edge connects a ligand to its scaffold, and a `protein_in_cluster` edge connects a protein to a sequence-identity cluster. Additional edges encode ligand Tanimoto similarity, pocket similarity, common assay membership, common publication, common data source, and shared decoy protocol.

Each edge type $r \in \mathcal{R}$ is assigned a leakage weight $w_r \in [0, 1]$, where larger values indicate stronger evidence of potential contamination. For instance, exact ligand identity receives a larger weight than scaffold similarity, and exact protein identity receives a larger weight than family-level similarity.

3.3 Path-Based Contamination Scoring

Given an example x_t and a reference set $\mathcal{A}$, such as a training split, we define the contamination of x_t with respect to $\mathcal{A}$ as the strength of the strongest leakage path connecting it to any example in $\mathcal{A}$. Let $\mathcal{P}_{\leq L}(x_i, x_t)$ denote the set of typed paths of length at most L between $x_i \in \mathcal{A}$ and x_t in $\mathcal{G}$. The score of a path π is defined as

$$S(\pi) = \prod_{e \in \pi} w_r(e),$$

where $r(e)$ is the type of edge e . The overall contamination score is

$$C(x_t, \mathcal{A}) = \max_{x_i \in \mathcal{A}} \max_{\pi \in \mathcal{P}_{\leq L}(x_i, x_t)} S(\pi).$$

This score captures both exact and indirect contamination. For example, an exact ligand overlap corresponds to a short high-weight path, while scaffold overlap or protein-family overlap corresponds to a lower-weight but still meaningful path. In practice, we compute both a global contamination score and axis-specific scores:

$$C(x_t, \mathcal{A}) = \max\{C_{\text{ligand}}, C_{\text{scaffold}}, C_{\text{protein}}, C_{\text{pocket}}, C_{\text{assay}}, C_{\text{source}}, C_{\text{time}}\}.$$

This decomposition allows us to diagnose which form of contamination dominates each benchmark or split.

3.4 Forbidden Leakage Relations

To construct clean train/validation/test partitions, we define a set of forbidden leakage relations $\mathcal{F} \subset \mathcal{R}$. These are relations that are not allowed to connect examples across train, validation, and test partitions. The strictest setting includes exact ligand identity, scaffold identity, high ligand similarity, exact protein identity, sequence-cluster membership, pocket-cluster membership, assay overlap, publication overlap, source-specific decoy protocol overlap, and invalid temporal ordering.

Table 2: Forbidden relations used to construct VS-LEAKKG-clean splits.

Axis	Forbidden relation	Purpose
Ligand	Same InChIKey	Prevent exact compound memorization
Scaffold	Same Bemis–Murcko scaffold	Prevent scaffold-level memorization
Chemical similarity	Morgan Tanimoto above threshold	Prevent near-duplicate ligand leakage
Protein	Same UniProt or PDB chain	Prevent exact target leakage
Protein family	Same sequence cluster	Prevent homologous target leakage
Pocket	Same or highly similar pocket cluster	Prevent binding-site leakage
Assay	Same assay identifier	Prevent assay-specific leakage
Publication	Same source document	Prevent publication-level leakage
Source/protocol	Same artificial decoy protocol	Prevent benchmark-source shortcuts
Time	Test timestamp earlier than train timestamp	Prevent temporal leakage

The choice of $\mathcal{F}$ defines the evaluation regime. A scaffold-clean split forbids scaffold relations; a protein-clean split forbids protein-cluster relations; a strict-clean split forbids all high-confidence relations in Table 2.

3.5 Leakage Group Construction

Given the full graph $\mathcal{G}$ and the forbidden relation set $\mathcal{F}$, we construct a forbidden subgraph

$$\mathcal{G}_{\mathcal{F}} = (\mathcal{V}, \mathcal{E}_{\mathcal{F}}), \quad \mathcal{E}_{\mathcal{F}} = \{e \in \mathcal{E} : r(e) \in \mathcal{F}\}.$$

We then compute connected components over benchmark examples induced by $\mathcal{G}_{\mathcal{F}}$. Each connected component defines a leakage group

$$g_m = \{x_i : x_i \text{ is connected to other examples through forbidden paths}\}.$$

All examples in the same leakage group must be assigned to the same partition. This prevents a scaffold, protein cluster, pocket cluster, assay, or source-specific group from being split across training and testing.

This group-level construction is the key step that converts graph contamination into a practical clean split. Rather than checking only pairwise exact duplicates after splitting, VS-LEAKKG prevents leakage before model training by ensuring that entire leakage groups are assigned atomically.

3.6 Leakage-Controlled Train/Validation/Test Split

Let $\mathcal{G}_{\text{grp}} = \{g_1, \dots, g_M\}$ denote the leakage groups. We assign each group to one of three partitions:

$$z_m \in \{\text{train}, \text{val}, \text{test}\}.$$

The resulting example-level split is obtained by assigning all $x_i \in g_m$ to z_m .

We formulate the group assignment as a constrained optimization problem:

$$\min_z \sum_{\substack{g_m, g_n \in \mathcal{G}_{\text{grp}} \\ z_m = \text{train}, z_n \in \{\text{val}, \text{test}\}}} \sum_{x_i \in g_m} \sum_{x_j \in g_n} C(x_j, \{x_i\}),$$

subject to label balance and benchmark-utility constraints:

$$\left| \frac{|\mathcal{D}_k^+|}{|\mathcal{D}_k|} - \frac{|\mathcal{D}^+|}{|\mathcal{D}|} \right| \leq \epsilon, \quad k \in \{\text{train}, \text{val}, \text{test}\},$$

where $\mathcal{D}^+$ denotes active examples. Additional constraints enforce minimum numbers of targets, ligands, active compounds, inactive compounds, and screening candidates in validation and test.

In practice, we use a two-stage procedure. First, leakage groups are constructed by connected components under $\mathcal{F}$. Second, groups are assigned to train/validation/test using a greedy or integer-programming-based heuristic that preserves the desired split ratio while minimizing residual cross-partition contamination. We explicitly report residual contamination after splitting, rather than claiming that the split is leakage-free in an absolute sense.

The resulting splits are released under multiple regimes:

- **Ligand-clean:** exact ligand overlap is forbidden across partitions.
- **Scaffold-clean:** scaffold overlap is forbidden across partitions.
- **Protein-clean:** proteins from the same sequence cluster are assigned to the same partition.
- **Pocket-clean:** similar pocket clusters are assigned to the same partition.
- **Assay-clean:** examples from the same assay or publication are assigned to the same partition.
- **Dual-clean:** both ligand/scaffold and protein/pocket contamination are controlled.
- **Strict-clean:** ligand, scaffold, protein, pocket, assay, source, protocol, and temporal contamination are controlled simultaneously.

3.7 Clean-Split Model Retraining Protocol

After constructing VS-LEAKKG-clean partitions, we retrain protein–ligand retrieval models using the same split for all methods. For a model f_θ , training is performed only on $\mathcal{D}_{\text{train}}$:

$$\theta^* = \arg \min_{\theta} \mathcal{L}(f_\theta; \mathcal{D}_{\text{train}}).$$

Hyperparameters and checkpoints are selected using only the leakage-controlled validation split:

$$\hat{\theta} = \arg \max_{\theta \in \Theta} M(f_\theta; \mathcal{D}_{\text{val}}),$$

where M is a validation metric such as BEDROC or $\text{EF}_{1\%}$. Final results are reported once on $\mathcal{D}_{\text{test}}$:

$$\text{Score} = M(f_{\hat{\theta}}; \mathcal{D}_{\text{test}}).$$

This protocol prevents leakage not only in final testing, but also in model selection. We therefore distinguish two failure modes: training contamination, where test-like examples are present in the training set, and validation contamination, where checkpoint selection is biased by validation examples that are too close to training or test examples.

For models that use both structure-based and ligand-based data, such as unified protein–ligand retrieval models, both data streams are filtered through the same VS-LEAKKG split. Structure-based complexes are assigned according to their protein, pocket, and ligand groups; ligand-based assay records are assigned according to assay, ligand, scaffold, protein, publication, and temporal groups. This ensures that pretraining and fine-tuning data do not leak into validation or test through different data modalities.

3.8 Shortcut Diagnostic Baselines

A key use of VS-LEAKKG is to test whether benchmark performance can be achieved without modeling protein–ligand interactions. We introduce four diagnostic baselines.

Ligand-only baseline. This baseline uses only ligand fingerprints, scaffold identity, or physicochemical descriptors to classify active and inactive compounds. Strong performance from this baseline indicates that the benchmark contains ligand-side or decoy-selection artifacts.

Source-only baseline. This baseline uses only dataset source, assay type, publication, or decoy protocol. Strong performance indicates source-specific label artifacts.

Contamination nearest-neighbor baseline. For a test example x_t , this baseline finds the most contaminated training example:

$$x^* = \arg \max_{x_i \in \mathcal{D}_{\text{train}}} C(x_t, \{x_i\}),$$

and transfers its label or ranking score to x_t . If this baseline performs competitively with deep protein–ligand models, then relational leakage alone may explain a substantial fraction of benchmark performance.

Dummy-receptor baseline. This baseline replaces the true protein or pocket with a fixed dummy receptor representation while preserving ligand information. Strong performance indicates that the model does not rely on target-specific structural information.

3.9 Evaluation Protocol

We evaluate models under both legacy and clean-split regimes. Given a query protein or pocket, each model ranks candidate ligands by a predicted score. We report standard virtual screening metrics:

$$\text{AUROC}, \quad \text{BEDROC}, \quad \text{EF}_{0.5\%}, \quad \text{EF}_{1\%}, \quad \text{EF}_{5\%}.$$

In addition, we report contamination-aware metrics:

- contamination distribution before and after clean split construction;
- performance by contamination decile;
- worst-group performance over scaffold, protein-family, pocket-cluster, assay, and source groups;
- model ranking changes from legacy evaluation to strict-clean retraining;
- relative gap between interaction-based models and shortcut baselines;
- validation-leakage effect, measured by comparing checkpoint selection on leaked versus clean validation splits.

This evaluation distinguishes four possible outcomes. First, if a model performs well only in high-contamination regimes, its apparent success may be driven by memorization or shortcut learning. Second, if ligand-only or source-only baselines remain competitive, the benchmark may not require structure-based interaction modeling. Third, if a model performs well on legacy evaluation but drops substantially after clean-split retraining, then the legacy benchmark likely overestimates generalization. Fourth, if a model maintains strong performance under strict-clean splits and consistently outperforms shortcut baselines, this provides stronger evidence of genuine protein–ligand generalization.

3.10 Optional: Contamination-Aware Training

Although the primary goal of VS-LEAKKG is benchmark governance and clean evaluation, the graph can also guide model training. We define contamination-aware contrastive learning by selecting hard negatives using graph relations. For a positive protein–ligand pair (b_i, ℓ_i) , hard negatives are selected from ligands that are chemically similar but experimentally inactive, or from ligands connected through scaffolds but not known to bind the same pocket.

A general training objective can be written as

$$\mathcal{L} = \mathcal{L}_{\text{contrastive}} + \lambda_1 \mathcal{L}_{\text{source-adv}} + \lambda_2 \mathcal{L}_{\text{groupDRO}} + \lambda_3 \mathcal{L}_{\text{KG-hard}},$$

where $\mathcal{L}_{\text{source-adv}}$ discourages the representation from encoding dataset-source shortcuts, $\mathcal{L}_{\text{groupDRO}}$ improves worst-group performance across scaffold, protein, pocket, and assay groups, and $\mathcal{L}_{\text{KG-hard}}$ encourages discrimination among graph-selected hard negatives. This extension is optional; the core contribution of VS-LEAKKG remains the contamination-aware benchmark construction and clean-split retraining framework.

4 Discussion

The proposed framework differs from existing virtual screening benchmarks in both goal and representation. DUD-E, DEKOIS, LIT-PCBA, and BayesBind provide specific benchmark datasets, while VS-LEAKKG provides a general mechanism for auditing, constructing, and retraining on benchmark splits. DrugOOD and DataSAIL address out-of-distribution evaluation and leakage-reduced splitting in broader biological machine learning settings, but they do not explicitly model the typed contamination pathways that arise in structure-based virtual screening. Biomedical knowledge graphs such as Hetionet and PrimeKG integrate biological knowledge for prediction and reasoning, whereas VS-LEAKKG uses a heterogeneous graph as a benchmark-integrity object.

This distinction is important because virtual screening contamination is inherently multi-relational. A benchmark can be clean along one axis while contaminated along another. For example, a test protein may be absent from training but belong to a highly similar pocket cluster; a test ligand may be unique but share a scaffold with many training ligands; or a target-fishing experiment may be held out as a dataset while many of its ligands already appear in ChEMBL. By exposing these paths explicitly, VS-LEAKKG enables more transparent and reliable assessment of progress in virtual screening.

The clean-split retraining protocol is equally important. Auditing existing benchmarks can reveal potential contamination, but retraining under graph-controlled partitions tests whether models continue to generalize once those pathways are removed. This turns VS-LEAKKG from a diagnostic tool into a reusable benchmark construction framework. The resulting evaluation does not claim to remove all possible sources of leakage; instead, it makes the controlled axes explicit, quantifies residual contamination, and provides a reproducible basis for comparing virtual screening models under well-defined cold-start assumptions.

5 Conclusion

We presented VS-LEAKKG, a contamination-aware heterogeneous graph framework for reliable virtual screening evaluation and clean benchmark construction. The framework models train-validation-test contamination as typed paths across ligands, scaffolds, proteins, pockets, assays, sources, decoy protocols, and time. It supports example-level contamination scoring, leakage-group construction, multi-axis train/validation/test splitting, shortcut diagnostics, and clean-split retraining of modern protein-ligand retrieval models. By shifting the focus from single benchmark scores to benchmark integrity and leakage-controlled retraining, VS-LEAKKG aims to provide a more reliable basis for measuring progress in structure-based virtual screening.

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